

ECE330 – Spring 2024
Exam 1
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90 min

Thursday, February 22, 2024

NAME = KEY =

NetID

Instructions:

- Clearly PRINT your name in CAPITAL LETTERS.
- Clearly write your NetID.
- This exam is closed book and closed notes.
- Please SHOW ALL calculations and steps to receive full credit.
 - Writing just the correct numerical answer with no derivation will not result in credit.
 - Writing just the formula will not result in partial credit.
- You may write on both sides of the exam paper. If there is anything important on the reverse side, please indicate this on the front side.
- All answers must INCLUDE UNITS whenever appropriate.
- Graphing and/or 4-function calculators are permitted, but no cellphones/laptops.
- This exam booklet contains 9 pages (including this one) with 4 problems.

Problem 1 (25 Points)

A single-phase source serves two loads connected in parallel.

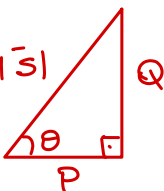
- Load 1 draws 180 W at a lagging PF of 0.6.
- Load 2 draws 1 A at a leading PF of 0.6.

The voltage across the loads is $\bar{V} = 100 \angle 30^\circ$ V. (Note that the voltage angle is not zero.)

a) Compute the current phasor $\bar{I}_1$ for the first load (180 W at a lagging PF of 0.6). Your answer should be a complex number, in either rectangular or polar form, whichever is more convenient.

$$\bar{I}_1 = \underline{3 \angle -23.13^\circ} = 2.959 - j1.179 \quad \text{Unit} = \underline{A}$$

We know $\bar{S} = \bar{V} \bar{I}^*$, $\bar{V} = 100 \angle 30^\circ$ V $\Rightarrow$ We need to find $\bar{S}$ to calculate $\bar{I}$.



$$\bar{S}_1 = \frac{P}{\text{pf}} \angle \cos^{-1}(\text{pf}) = \frac{180}{0.6} \angle \cos^{-1}(0.6) = 300 \angle 53.13^\circ \text{ VA}$$

"lagging", $\theta = \cos^{-1}(\text{pf})$

$$\text{Then, } \bar{I}_1 = \left(\frac{\bar{S}_1}{\bar{V}} \right)^* = \frac{300 \angle -53.13^\circ}{100 \angle -30^\circ} = 3 \angle -23.13^\circ \text{ A}$$

b) Compute the current phasor $\bar{I}_2$ for the second load (1 A at a leading PF of 0.6). Your answer should be a complex number, in either rectangular or polar form, whichever is more convenient.

$$\bar{I}_2 = \underline{1 \angle 83.13^\circ} \quad \text{Unit} = \underline{A}$$

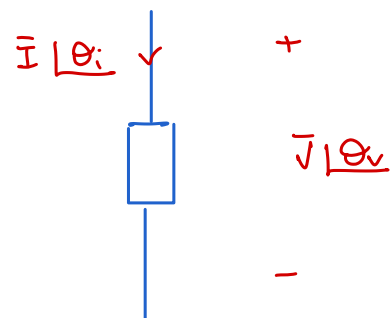
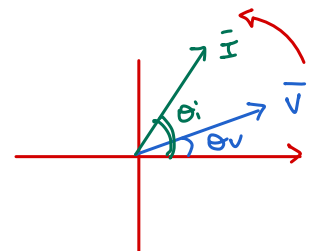
$$\text{pf} = 0.6, \text{ leading} \Rightarrow \cos(\theta_v - \theta_i) = 0.6$$

$$\theta_v - \theta_i = 53.13^\circ \quad \times$$

$$\text{OR, } \theta_v - \theta_i = -53.13^\circ \quad \checkmark \rightarrow \text{pf is leading; Current phasor leads the voltage phasor}$$

$$\Rightarrow \theta_i = 30^\circ + 53.13^\circ = 83.13^\circ$$

$$\Rightarrow \bar{I}_2 = 1 \angle 83.13^\circ \text{ A}$$



c) How many VARs of capacitance do we need to make the overall PF unity? Your answer should be stated as a numerical value.

Capacitance= +160 Unit= VAR

First, find $\bar{S}_{tot} = \bar{S}_1 + \bar{S}_2$

$$\bar{S}_2 = \bar{V} \cdot \bar{I}_2^* = 100 \angle 30^\circ \times 1 \angle -83.13^\circ = 100 \angle -53.13^\circ \text{ VA}$$

$$\bar{S}_{tot} = \underbrace{300 \angle 53.13^\circ}_{= \bar{S}_1 \text{ from part a}} + 100 \angle -53.13^\circ = 288.444 \angle 33.69^\circ = 240 + j160 \text{ VA}$$

$\Rightarrow$ Then, we need to add -160 VAR reactive power to make the overall pf unity. Note that capacitance implies negative VAR.

Therefore, the answer is -160 VAR.

d) How many VARs of capacitance do we need to make the overall PF 0.8 leading? Your answer should be stated as a numerical value. Please indicate your reasoning for why your answer should correctly yield a leading and not lagging overall PF.

Capacitance= 340 Unit= VAR

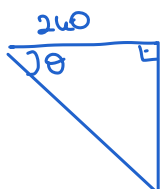
Reason why answer above gives overall leading (and not lagging)

Possible answers: $Q_{final} < 0$, $\theta_{load} < 0$ or the

final line current phasor leads the voltage phasor, or the power triangle is now flipped (showing the original or final power triangle). Note that listing the solution steps of part c & d is not an acceptable explanation here.

$$P_{tot} = 240 \text{ W}, \quad \theta = \cos^{-1}(0.8)$$

$\hookrightarrow$ since leading pf



$Q' = ?$

$$Q' = -P_{tot} \tan(\cos^{-1}(0.8)) = -240 \times \frac{3}{4} = -180 \text{ VAR}$$

$$Q_{tot}^{add} = -180 - 160 \text{ VAR} \Rightarrow \text{Capacitance} = |-340| \text{ VAR}$$

The reactive power we added in part c to make pf unity

Problem 2 (25 Points)

A 3-phase, 208 V (line-line), Wye-connected source is supplying two 3-phase loads connected in parallel:

1. Wye-connected, inductive, draws 10 A phase current and 3100 W of real power (3-phase)
2. Delta-connected, it has per-phase impedance $\bar{Z}_2$, draws 8 A line current and 1800 VARs of reactive power (3-phase) at a leading power factor.

a) Find the total real power (3-phase) supplied by the source.

$$P_{\text{tot}} = \underline{5350.931} \quad \text{Unit} = \underline{W}$$

$$P_{\text{tot}} = P_1 + P_2 \quad , \quad \text{where} \quad P_1 = 3100 \text{ W} \quad \text{given.}$$

$$\begin{aligned} \text{Load 2: } I_{2,\text{line}} &= 8 \text{ A} \quad , \quad V_{2,\text{ll}} = 208 \text{ V} \quad \Rightarrow \quad S_2 = \sqrt{3} \cdot V_{2,\text{ll}} I_{2,\text{line}} \\ &= \sqrt{3} \times 208 \times 8 \\ &= 2882.133 \text{ VA} \end{aligned}$$

$$P_2 = \sqrt{S_2^2 - Q_2^2} = \sqrt{2882.133^2 - 1800^2} = 2250.93 \text{ W}$$

$$P_{\text{tot}} = 3100 + 2250.93 = 5350.931 \text{ W}$$

b) Compute the total line current (magnitude) being supplied by the source.

$$I_L = \underline{14.853} \quad \text{Unit} = \underline{A}$$

$$\text{Find } S_{\text{tot}} \text{ , then } I_L = \frac{S_{\text{tot}}}{\sqrt{3} \cdot 208}$$

$$\text{Load 1: } S_1 = \sqrt{3} \cdot 208 \cdot 10 = 3602.666 \text{ VA} \quad (\text{Y-connected load; } I_{\text{phase}} = I_{\text{line}})$$

$$Q_1 = \mp \sqrt{S_1^2 - P_1^2} = \sqrt{3602.666^2 - 3100^2} = \mp 1835.539 \text{ VAR}$$

$$Q_1 = +1835.539 \text{ VAR} \quad \text{since inductive load.}$$

$$\bar{S}_{\text{tot}} = P_{\text{tot}} + j(Q_1 + Q_2) = 5350.931 + j(1835.539 \ominus 1800) = 5351.05 \angle 0.38^\circ$$

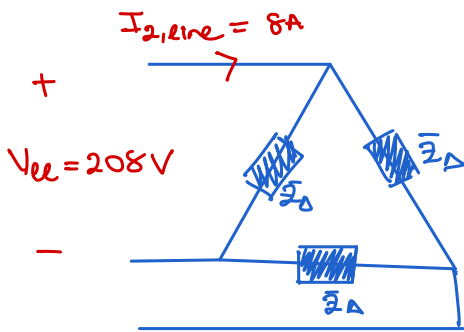
$$\text{Then, } I_L = \frac{5351.05}{\sqrt{3} \cdot 208} = 14.853 \text{ A}$$

Load 2 has leading pf!

c) Find per-phase impedance of the second load, $\bar{Z}_2$

$$\bar{Z}_2 = \underline{35.171 - j 28.125} = 45.033 \angle -38.65^\circ \text{ Unit} = \underline{\Omega}$$

Delta connected load: $V_{\text{phase}} = V_{LL}$, $\bar{S}_{2\phi} = \frac{1}{3} \bar{S}_2$



$$\bar{S}_{2,3\phi} = 2250.93 - j 1800 \text{ VA (from part a)}$$

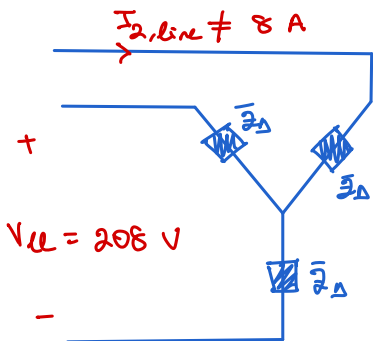
$$\bar{S}_{2,\phi} = \bar{V}_\phi \bar{I}_\phi^* = \bar{V}_\phi \left(\frac{\bar{V}_\phi}{\bar{Z}_\Delta} \right)^* = \frac{208^2}{\bar{Z}_\Delta^*}$$

$$\Rightarrow Z_\Delta = \frac{208^2}{(\bar{S}_2 / 3)} = \left(\frac{208^2 \times 3}{2250.93 - j 1800} \right)^* = 35.171 - j 28.125$$

d) Now we reconfigure the second load to be connected in Wye (instead of Delta, as originally indicated), while keeping its per-phase impedance $\bar{Z}_2$ the same as computed in part c. What is the new total line current (magnitude) being supplied by the source?

$$I_{L,\text{new}} = \underline{11.224} \text{ Unit} = \underline{A}$$

Y-connected load: $V_{\text{phase}} = \frac{V_{LL}}{\sqrt{3}}$, $S_{3\phi} = 3 S_{1\phi}$



$$* I_{2,\text{line}} \neq 8 \text{ A since } V_{LL} = 208 / \sqrt{3} .$$

$$\bar{S}_{2,3\phi} = 3 \times \frac{V_{\text{phase}}^2}{\bar{Z}_\Delta^*} = \frac{3 (208/\sqrt{3})^2}{45.033 \angle +38.65^\circ} = 960.718 \angle -38.65^\circ$$

$$\bar{S}_{\text{tot,new}} = \bar{S}_1 + \bar{S}_{2,\text{new}} = (3100 + j 1835.539) + 960.718 \angle -38.65^\circ = 4043.674 \angle 17.79^\circ \text{ VA}$$

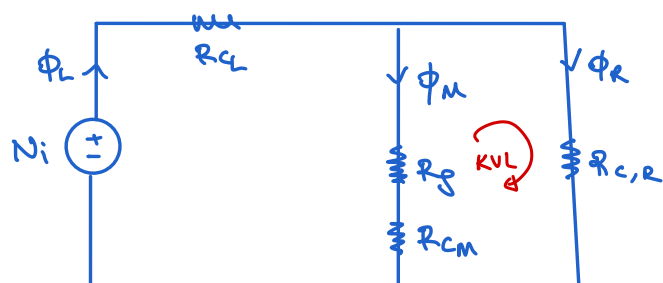
$$I_{L,\text{new}} = \frac{4043.674}{\sqrt{3} \cdot 208} = 11.224 \text{ A}$$

b) How much current i is needed to achieve a flux density of $B = 0.5$ T in the center leg?

$$i = \underline{285.761} \quad \text{Unit} = \underline{A}$$

We have $Ni = \phi_L R_{eq}$ and $\phi_L = \phi_M + \phi_R$

$$\Rightarrow i = \phi_L \frac{R_{eq}}{100}$$



KVL: $\phi_M (R_g + R_{c,M}) = \phi_R R_{c,R}$

$$\phi_M \frac{0.23}{11 \mu\Omega} = \phi_R \frac{0.33}{11 \mu\Omega} \Rightarrow \phi_R = 0.697$$

$\Rightarrow$ Since $\phi_L = \phi_R + \phi_M$, $\phi_L = 1.697 \phi_M$

For $B = 0.5$ T $\Rightarrow \phi_M = BA = 0.5 \times 6 \text{ cm}^2 = \underline{\underline{3 \cdot 10^{-4} \text{ Wb}}}$

$$\begin{aligned} \text{Then, } i &= \phi_L \frac{R_{eq}}{100} = 1.697 \phi_M \frac{56130537.97}{100} \\ &= 285.761 \text{ A} \end{aligned}$$

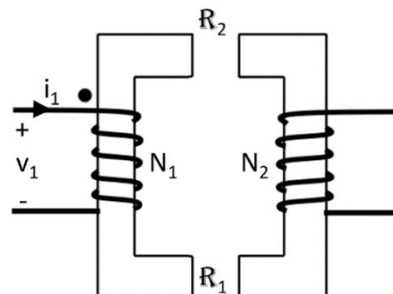
Problem 4 (25 Points)

Please show your calculations and steps to receive full credit.

Writing just the correct numerical answer with no derivation will not result in credit.

Writing just the formula will not result in partial credit.

a) Two coils are wrapped around an iron core as shown (not to scale). Coil 1 on the left has 10 turns, and Coil 2 on the right has 20 turns. The two air gaps have reluctances $R_1 = 5 \text{ At/Wb}$, $R_2 = 15 \text{ At/Wb}$. The reluctance of the core can be neglected.



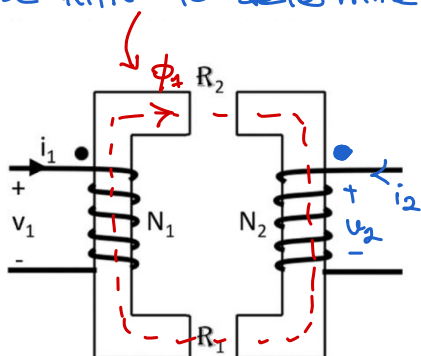
What is the self-inductance for each coil L_1 and L_2 and the mutual inductance between the two coils M ? (Recall that M should always be a positive number.)

$$L_1 = \underline{5} \text{ Unit} = \underline{H} \quad L_2 = \underline{20} \text{ Unit} = \underline{H}$$

$$M = \underline{10} \text{ Unit} = \underline{H}$$

* Dot convention: "Current entering dots throw flux in the same direction!"

- ① Select a coil and put a dot on one terminal arbitrary (Already done in the question.)
- ② Mark the dotted terminal with + voltage polarity and inject current into first dot. (Already done in question)
- ③ Use RHR to determine direction:

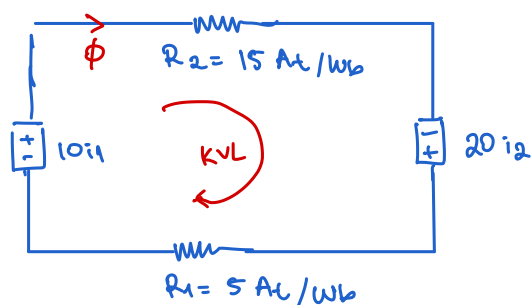


- ④ Determine current direction in the second winding which will throw flux in the same direction as the first winding.

- ⑤ Put the dot in the 2nd wdg, where current enters the terminal.

- ⑥ Put + voltage polarity on the 2nd dotted terminal.

Magnetic ckt:



$$\text{KVL} \quad 20\phi = 10i_1 + 20i_2$$

$$\lambda_1 = N_1 \phi = 10 (0.5 i_1 + i_2) = 5 i_1 + 10 i_2$$

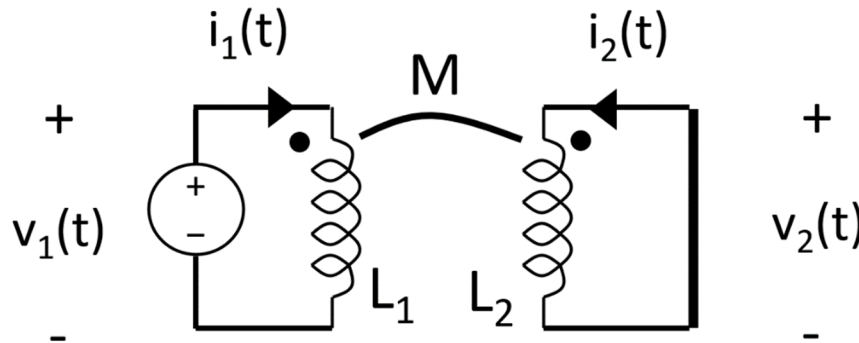
$$\lambda_2 = N_2 \phi = 20 (0.5 i_1 + i_2) = 10 i_1 + 20 i_2$$

$$L_1 = \frac{d\lambda_1}{di_1} = 5, \quad M = \frac{d\lambda_1}{di_2} = \frac{d\lambda_2}{di_1} = 10$$

$$L_2 = \frac{d\lambda_2}{di_2} = 20 \text{ H}$$

Part b) is unrelated to part a).

b) The coupled coils below have $L_1 = 10 \text{ H}$, $L_2 = 5 \text{ H}$ and $M = 5 \text{ H}$. The primary is driven by a voltage source and the secondary has been short-circuited.



If $v_1(t) = \sqrt{2} \cos(120\pi t)$ is applied to the first coil, what is the short-circuit current $i_2(t)$ observed at the second coil?

Please give an answer as a function of time. Please ensure that your answer is consistent with the polarity of the currents as indicated in the diagram.

$$i_2(t) = \frac{\sqrt{2}}{5 \times 120\pi} \cos(120\pi t + \pi/2) \quad \text{Unit} = \underline{A}$$

$$\begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix} = \begin{bmatrix} L_1 & M \\ M & L_2 \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_1(t) \\ i_2(t) \end{bmatrix}, \quad \text{Where } \begin{matrix} L_1 = 10 \text{ H} \\ M = 5 \text{ H} \\ L_2 = 5 \text{ H} \end{matrix}$$

$$(I) \quad v_1(t) = \sqrt{2} \cos(120\pi t) = 10 \frac{di_1(t)}{dt} + 5 \frac{di_2(t)}{dt}$$

$$(II) \quad v_2(t) = 0 = 5 \frac{di_1(t)}{dt} + 5 \frac{di_2(t)}{dt} \Rightarrow \frac{di_2(t)}{dt} = - \frac{di_1(t)}{dt}$$

$$\text{Plug into eq (I): } \sqrt{2} \cos(120\pi t) = -10 \frac{di_2(t)}{dt} + 5 \frac{di_2(t)}{dt} = -5 \frac{di_2(t)}{dt}$$

$$\int \frac{di_2(t)}{dt} dt = \int -\frac{\sqrt{2}}{5} \cos(120\pi t) dt \quad 0 \text{ since AC signal}$$

$$\text{Then, } i_2(t) = -\frac{\sqrt{2}}{5 \times 120\pi} \sin(\underbrace{120\pi t}_{:= \theta}) \quad +C$$

* Recall that $-\sin(\theta) = \sin(-\theta) = \sin(\pi + \theta) = \cos(\theta + \pi/2) = \dots$ All are acceptable.

$$i_2(t) = \frac{\sqrt{2}}{5 \times 120\pi} \cos(120\pi t + \pi/2) \quad A$$